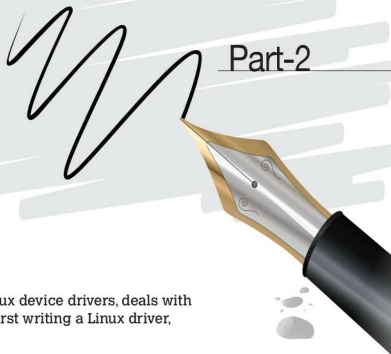


Writing Your First Linux Driver in the Classroom



This article, which is part of the series on Linux device drivers, deals with the concept of dynamically loading drivers, first writing a Linux driver, before building and then loading it.

Shweta and Pugs reached their classroom late, to find their professor already in the middle of a lecture. Shweta sheepishly asked for his permission to enter. An annoyed Professor Gopi responded, “Come on! You guys are late again; what is your excuse, today?” Pugs hurriedly replied that they had been discussing the very topic for that day’s class—device drivers in Linux. Pugs was more than happy when the professor said, “Good! Then explain dynamic loading in Linux. If you get it right, the two of you are excused!” Pugs knew that one way to make his professor happy was to criticise Windows.

He explained, “As we know, a typical driver installation on Windows needs a

reboot for it to get activated. That is really not acceptable; suppose we need to do it on a server? That’s where Linux wins. In Linux, we can load or unload a driver on the fly, and it is active for use instantly after loading. Also, it is instantly disabled when unloaded. This is called dynamic loading and unloading of drivers in Linux.”

This impressed the professor. “Okay! Take your seats, but make sure you are not late again.” The professor continued to the class, “Now you already know what is meant by dynamic loading and unloading of drivers, so I’ll show you how to do it, before we move on to write our first Linux driver.”

Dynamically loading drivers

These dynamically loadable drivers are more